

Sophie Rollins

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Research Interests

My research lies in designing experiences that foster technological literacy in citizens in order to inform and engage them in critical conversations about the growth and ethics of new technologies.

Education

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| 2024 - Present (expected 2029) | PhD student, Technology and Social Behavior
Joint program in Computer Science and Communication Studies
Advisor: Prof. Duri Long
Northwestern University, Evanston, IL |
| 2023 | Master of Science, Computer Science
Northwestern University, Evanston, IL |
| 2023 | Bachelor of Science, Computer Science
Northwestern University, Evanston, IL |

Publications

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| 2026 | Governing Together: Toward Infrastructure for Community-Run Social Media
Sohyeon Hwang, Sophie Rollins* , Thatiany Andrade Nunes*, Yuhan Liu, Richmond Y. Wong, Aaron Shaw, Andrés Monroy-Hernández.
CHI '26: Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.
* authors contributed equally |
| 2026 | Designing and Evaluating Museum Exhibit Prototypes to Foster Middle Schoolers' AI Literacy through Creativity and Embodiment
Hasti Darabipourshiraz*, Sophie Rollins* , Milka Trajkova, Chengzhi Zhang, Yasmine Belghith, Tom McKlin, Jessica Roberts, Brian Magerko, Duri Long.
TEI '26: Proceedings of the 20th International Conference on Tangible, Embedded, and Embodied Interaction.
* authors contributed equally |
| 2026 | AI Unplugged: Exploring Pathways from Physical Simulation to Conceptualization of AI Reasoning Processes
Hasti Darabipourshiraz, Lily Murakami Ng, Grace Wang, Sophie Rollins , Duri Long.
ACM Transactions on Computing Education. |

- 2024 **Knowledge Net: Fostering Children's Understanding of Knowledge Representations Through Creative Making and Embodied Interaction in a Museum Exhibit**
Sophie Rollins, Katherine Hancock, Jasmin Ali-Diaz, Nyssa Shahdadpuri, Duri Long.
 C&C '24: Proceedings of the 16th Conference on Creativity & Cognition.
- 2023 **Fostering AI Literacy with Embodiment & Creativity: From Activity Boxes to Museum Exhibits**
 Duri Long, **Sophie Rollins**, Jasmin Ali-Diaz, Katherine Hancock, Samnang Nuonsinoeun, Jessica Roberts, Brian Magerko.
 IDC '23: Proceedings of the 22nd Annual ACM Interaction Design and Children Conference.

Honors

- 2025-2026 Design Research Cluster Fellowship (Northwestern University Center for Human-Computer Interaction + Design)
- 2024-2025 Jaharis Family Fellowship (Northwestern University Department of Communication Studies)
- 2024-2025 First-Year Cognitive Science Fellowship (Northwestern University Department of Cognitive Science)

Research Experience

- 2023 – 2024 **Pre-Doctoral Scholar, Creative Interfaces Research + Design Studio**
 Principal Investigator: Professor Duri Long, Northwestern University
Museum Exhibits to Foster AI Literacy
 Developed a web application for two museum exhibits aimed at fostering AI literacy in middle school youth by teaching about knowledge representations and machine learning through embodied interactions. Built and trained a data-to-text NLG model to transform a semantic network into a story in order to demonstrate the possible creative outputs generated from knowledge representations.
- Computational Embodiments*
 Researched and designed knowledge representations of the human body outside of the traditional skeletal model. Designed a deep motion editing tool which allows users to record a dance and visualize what that dance would look like when performed with varying levels of tension.

Teaching Experience

- 2022 **Peer Mentor, Introduction to Web Development**
 Northwestern University, Evanston, IL
- 2022 **Peer Mentor, Generative Methods**
 Northwestern University, Evanston, IL

Industry Experience

2022

Software Engineering Intern

Lyft, Inc., Remote

Improved communication surrounding the rider “rematch” experience through implementation of new messages and notifications that provide context to Lyft riders when they experience a driver cancellation or change. Collaborated with project managers, data scientists, and other software engineers to deploy a production-scale user experiment on the Lyft app in order to measure the impact of the new communications.

2020-2021

Software Engineering Intern

RightHand Technologies, Inc., Chicago, IL

Collaborated with team members and users to develop the RightHand Intranet System, a web application utilized internally by RightHand Technologies employees. Centralized data collection for RightHand Technologies by consolidating data collection forms, a timecard system, company calendars, a clock-in system, and other features vital to company operations into one comprehensive website.

Mentorship and Service

2020 – 2023

Peer Adviser, New Student Experiences

Northwestern University, Evanston, IL

Served as year-long mentor for groups of engineering students in the Northwestern classes of 2024, 2025, and 2026. Led McCormick First-Year Experience personal development course for engineering first-year students, teaching academic and professional success, as well as personal well-being.

2022 – 2023

Program Director, Society of Women Engineers

Northwestern University, Evanston, IL

Organized and promoted Northwestern SWE Industry Day, Northwestern’s largest engineering career fair. Led the Professional Development committee, which hosts events to guide and advise undergraduate engineering students in gender minorities as they explore career opportunities.

Skills

Technical

Python, JavaScript, HTML/CSS, SQL, Git, Node.js, Microsoft Excel, MATLAB, machine learning (NumPy, Pandas, SciPy, scikit-learn, jupyter), data visualization (Tableau, Matplotlib).

Languages

English (fluent), Spanish (advanced).

Other Interests

Fibre arts (crochet, knitting, embroidery).